

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I K Phani Sridhar Yogi 192311091 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. SSL/TLS Overhead and Web Server Performance Optimization

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:

- A. Calculate the total data transmitted per second including TLS overhead.
 - B. Determine the percentage overhead due to TLS encryption.
 - C. Estimate bandwidth consumed due to handshake overhead per second.
 - D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).
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2. PGP-Based Email Encryption Load in Enterprise System

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:

- A. Calculate the total data transmitted per day including overhead.
 - B. Determine the percentage increase in storage due to PGP.
 - C. Analyze the computational overhead during encryption/decryption.
 - D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).
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3. IPSec Transport vs Tunnel Mode Performance Comparison

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:

- A. Compute total packet size for both modes.
 - B. Calculate bandwidth usage for each mode per second.
 - C. Compare efficiency and security benefits of both modes.
 - D. Recommend the appropriate mode for internal vs external communication with justification.
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4. S/MIME Certificate Management in Large Organization

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:

- A. Calculate total storage required during the renewal period.
 - B. Estimate certificate distribution overhead across the network.
 - C. Analyze challenges in certificate lifecycle management at scale.
 - D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).
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5. Email Encryption Latency in Cloud-Based Systems

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:

- A. Calculate total encryption time per email.
- B. Estimate total processing time per hour.
- C. Analyze the performance bottleneck between symmetric and asymmetric operations.
- D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

Question 1: SSL/TLS Overhead and Web Server Performance Optimization**Given Data**

- HTTP payload per response = 2000 bytes
- TLS overhead per response = 25 bytes
- Requests per second = 10,000
- New TLS handshakes per second = 1,000
- Handshake overhead per session = 3 KB = 3000 bytes

A. Total Data Transmitted per Second Including TLS Overhead

Total size of one HTTP response:

$$[2000 + 25 = 2025 \text{ bytes}]$$

For 10,000 requests per second:

$$[2025 \times 10000 = 20,250,000 \text{ bytes/second}]$$

Therefore,

$$[\boxed{20,250,000 \text{ bytes/second}}]$$

or approximately:

$$[\boxed{20.25 \text{ MB/s}}]$$

B. Percentage Overhead Due to TLS Encryption

TLS overhead per response = 25 bytes.

$$[\text{Percentage Overhead}]$$
$$[\frac{25}{2000} \times 100]$$

$$[= 1.25\%]$$

Therefore,

$$[\boxed{1.25\%}]$$

The TLS encryption adds only 1.25% overhead to the HTTP payload.

C. Bandwidth Consumed Due to Handshake Overhead

Handshake overhead per session = 3 KB = 3000 bytes.

New handshakes per second = 1000.

$$[3000 \times 1000]$$
$$[3,000,000 \text{ bytes/second}]$$

Therefore, the bandwidth consumed by handshake overhead is:

$$[\boxed{3,000,000 \text{ bytes/second}}]$$

or approximately:

[
\\boxed{3\\text{ MB/s}}
]

D. Security and Latency Trade-Off

TLS provides important security features such as confidentiality, authentication, and integrity. However, establishing a new TLS session requires additional processing and communication, which can increase latency and consume bandwidth.

In this system, the per-response TLS overhead is relatively small at 1.25%, but the 1,000 new handshakes per second introduce an additional 3 MB/s of overhead.

Optimization Techniques

1. **Session Resumption:** Reuse previously established TLS sessions instead of performing a complete handshake for every new connection. This reduces handshake processing and latency.
2. **HTTP/2:** HTTP/2 allows multiple requests and responses to share a single connection, reducing the number of connections and TLS handshakes required.
3. **QUIC:** QUIC integrates TLS 1.3 and provides faster connection establishment while reducing latency. It is used by HTTP/3.
4. **Connection Reuse:** Persistent connections can reduce the frequency of TLS handshakes and improve server performance.

Conclusion

TLS provides strong security with relatively low per-response overhead. However, frequent new TLS handshakes can significantly increase bandwidth consumption and latency. Using session resumption, HTTP/2, connection reuse, and QUIC can reduce this overhead while maintaining secure communication.

Question 2: PGP-Based Email Encryption Load in Enterprise System

Given Data

- Attachment size per email = 10 MB
- PGP overhead per email = 800 bytes
- Number of emails processed per day = 5,000

A. Total Data Transmitted per Day Including PGP Overhead

Assuming:

[
1\\text{ MB}=1,000,000\\text{ bytes}
]

Attachment size:

[
10\\text{ MB}=10,000,000\\text{ bytes}
]

Total size of one email:

$$[10,000,000 + 800 = 10,000,800 \text{ bytes}]$$

For 5,000 emails:

$$[10,000,800 \times 5,000]$$

$$[= 50,004,000,000 \text{ bytes/day}]$$

Therefore, the total data transmitted per day is:

$$[\boxed{50,004,000,000 \text{ bytes/day}}]$$

or approximately:

$$[\boxed{50.004 \text{ GB/day}}]$$

B. Percentage Increase in Storage Due to PGP

Total original data:

$$[10,000,000 \times 5,000]$$

$$50,000,000,000 \text{ bytes}$$

Total PGP overhead:

$$[800 \times 5,000]$$

$$4,000,000 \text{ bytes}$$

Percentage increase:

$$[\frac{4,000,000}{50,000,000,000} \times 100]$$

$$[= 0.008\%]$$

Therefore:

$$[\boxed{0.008\%}]$$

The storage increase caused by PGP overhead is very small.

C. Computational Overhead During Encryption and Decryption

PGP uses both symmetric and asymmetric cryptographic techniques. The large email attachment is generally encrypted using symmetric encryption because it is

computationally efficient for large amounts of data. Public-key cryptography is used for protecting the encryption key and for digital signatures.

The main computational overhead comes from:

1. Encryption of the 10 MB attachment.
2. Decryption of the encrypted attachment.
3. Public-key operations for key protection.
4. Digital signature generation and verification.
5. Hash calculation for integrity verification.

Although these operations require additional CPU resources, the 800-byte PGP overhead is very small compared with the 10 MB attachment.

D. Optimization Strategies

1. Compression

Compressing attachments before encryption can reduce the amount of data that needs to be encrypted and transmitted.

2. Batching

Processing multiple email encryption operations together can reduce repeated processing overhead and improve system throughput.

3. Hardware Acceleration

Cryptographic hardware acceleration can speed up encryption and decryption operations and reduce CPU usage.

4. Efficient Key Management

Efficient key management reduces unnecessary public-key operations and improves the overall performance of the email system.

Conclusion

PGP introduces only a small storage and transmission overhead of 0.008% for the given workload. However, encryption, decryption, key management, and digital signature operations require additional computational resources. Compression, batching, and hardware acceleration can improve the performance of a large-scale enterprise email system.

Question 3: IPSec Transport vs Tunnel Mode Performance Comparison

Given Data

- Original payload = 1000 bytes
- Original IP header = 20 bytes
- Transport Mode overhead = 30 bytes
- Tunnel Mode overhead = 50 bytes
- Traffic volume = 20,000 packets/second

A. Total Packet Size for Both Modes

Transport Mode

Original packet size:

[
 $1000 + 20 = 1020 \text{ bytes}$
]
Transport Mode adds 30 bytes:
[
 $1020 + 30 = 1050 \text{ bytes}$
]
Therefore:
[
 $\boxed{1050 \text{ bytes}}$
]

Tunnel Mode

Original packet size:

[
 $1000 + 20 = 1020 \text{ bytes}$
]
Tunnel Mode adds 50 bytes:
[
 $1020 + 50 = 1070 \text{ bytes}$
]
Therefore:
[
 $\boxed{1070 \text{ bytes}}$
]

B. Bandwidth Usage per Second

Transport Mode

[
 $1050 \times 20,000$
 $21,000,000 \text{ bytes/second}$
]
Therefore:
[
 $\boxed{21 \text{ MB/s}}$
]

Tunnel Mode

[
 $1070 \times 20,000$
 $21,400,000 \text{ bytes/second}$
]
Therefore:
[
 $\boxed{21.4 \text{ MB/s}}$
]

Comparison

Tunnel Mode consumes:

[
21,400,000-21,000,000
400,000\text{ bytes/second}
]

more bandwidth than Transport Mode.

C. Efficiency and Security Comparison

Feature	Transport Mode	Tunnel Mode
Packet Size	1050 bytes	1070 bytes
Bandwidth Usage	21 MB/s	21.4 MB/s
Overhead	30 bytes	50 bytes
Efficiency	Higher	Slightly lower
Security	Protects the IP payload	Protects the entire original IP packet
Typical Usage	Host-to-host communication	VPN and gateway-to-gateway communication

Transport Mode is more bandwidth-efficient because it adds only 30 bytes of overhead. Tunnel Mode has higher overhead but provides stronger protection for the original IP packet.

D. Recommended Mode

Internal Communication

For internal host-to-host communication, Transport Mode is recommended because it provides security with lower overhead and better bandwidth efficiency.

External Communication

For communication over external or untrusted networks, Tunnel Mode is recommended because it encrypts and encapsulates the complete original IP packet. It also provides better protection for the original IP addressing information.

Conclusion

Transport Mode produces a packet size of 1050 bytes and consumes 21 MB/s, while Tunnel Mode produces a packet size of 1070 bytes and consumes 21.4 MB/s. Transport Mode is more efficient, whereas Tunnel Mode provides greater protection and is more suitable for external network and VPN communication.

Question 4: S/MIME Certificate Management in Large Organization

Given Data

- Number of employees = 12,000
- Size of each certificate = 2 KB
- Renewal period = 1 year
- During renewal, old and new certificates are stored simultaneously for 1 month

A. Total Storage Required During the Renewal Period

During the renewal period, both the old and new certificates are stored.

Therefore, each employee requires:

$$[2 \times 2 = 4 \text{ KB}]$$

For 12,000 employees:

$$[12,000 \times 4 \\ 48,000 \text{ KB}]$$

Converting to MB:

$$[\frac{48,000}{1024} \\ 46.875 \text{ MB}]$$

Therefore, the total storage required is:

$$[\boxed{46.875 \text{ MB}}]$$

B. Certificate Distribution Overhead Across the Network

The size of one certificate is:

$$[2 \text{ KB}]$$

For 12,000 employees:

$$[12,000 \times 2 \\ 24,000 \text{ KB}]$$

Converting to MB:

$$[\frac{24,000}{1024} \\ 23.4375 \text{ MB}]$$

Since both old and new certificates are involved during renewal:

$$[23.4375 \times 2 \\ 46.875 \text{ MB}]$$

Therefore, the certificate distribution overhead during the renewal period is approximately:

$$[\boxed{46.875 \text{ MB}}]$$

C. Challenges in Certificate Lifecycle Management

Managing 12,000 S/MIME certificates creates several challenges:

- 1. Certificate Renewal:** Certificates must be renewed before expiration to prevent disruption of secure email communication.
- 2. Certificate Distribution:** Certificates need to be securely distributed to all employees.
- 3. Key Management:** Private keys must be protected from unauthorized access.
- 4. Certificate Revocation:** Compromised or invalid certificates must be revoked quickly.
- 5. Tracking and Monitoring:** The organization must maintain records of certificate status, expiration dates, and renewals.
- 6. Scalability:** Manual certificate management becomes difficult and error-prone when the number of employees is large.

D. Optimization Strategies

1. Automation

Automating certificate issuance, renewal, and revocation can significantly reduce administrative work and human errors.

2. Centralized PKI

A centralized Public Key Infrastructure (PKI) can manage certificates, keys, policies, and certificate authorities from a single system.

3. Short-Lived Certificates

Using short-lived certificates can reduce the security risks associated with compromised or outdated certificates.

4. Centralized Monitoring

A certificate management system can monitor expiration dates and automatically generate renewal alerts.

Conclusion

For 12,000 employees, the organization requires approximately 46.875 MB of certificate storage during the renewal period when both old and new certificates are retained. At this scale, automated certificate management and centralized PKI are recommended to simplify certificate distribution, renewal, monitoring, and revocation.

Question 5: Email Encryption Latency in Cloud-Based Systems

Given Data

- AES encryption time = 0.2 ms per MB
- RSA key exchange time = 5 ms per email
- Attachment size = 8 MB
- Emails processed = 8,000 emails per hour

A. Total Encryption Time per Email

AES encryption time for an 8 MB attachment:

$$[8 \times 0.2 = 1.6 \text{ ms}]$$

RSA key exchange time:

$$[5 \text{ ms}]$$

Therefore, total encryption time per email is:

$$[1.6 + 5 = 6.6 \text{ ms}]$$

Hence:

$$[\boxed{6.6 \text{ ms per email}}]$$

B. Total Processing Time per Hour

Number of emails processed per hour:

$$[8,000]$$

Encryption time per email:

$$[6.6 \text{ ms}]$$

Total processing time:

$$[8,000 \times 6.6 \\ 52,800 \text{ ms}]$$

Converting milliseconds to seconds:

$$[\frac{52,800}{1000} \\ 52.8 \text{ seconds}]$$

Therefore:

$$[\boxed{52.8 \text{ seconds per hour}}]$$

C. Performance Bottleneck

AES encryption takes:

$$[1.6 \text{ ms}]$$

RSA key exchange takes:

[
5\text{ ms}

]

Total:

[

6.6\text{ ms}

]

Percentage contribution of AES:

[

$\frac{1.6}{6.6} \times 100$

$\approx 24.24\%$

]

Percentage contribution of RSA:

[

$\frac{5}{6.6} \times 100$

$\approx 75.76\%$

]

Therefore, RSA key exchange is the main performance bottleneck, contributing approximately 75.76% of the total encryption time per email.

D. Optimization Strategies

1. ECC Adoption

Elliptic Curve Cryptography (ECC) can provide comparable security with smaller key sizes and lower computational requirements than traditional RSA in many applications.

2. Parallel Processing

Multiple emails can be encrypted simultaneously using multiple CPU cores or distributed processing, reducing the total processing time.

3. Hardware Acceleration

Dedicated cryptographic hardware or CPU cryptographic acceleration can improve AES performance and reduce CPU utilization.

4. Efficient Key Management

Session or symmetric keys can be reused appropriately to reduce repeated expensive asymmetric operations.

Conclusion

The total encryption time for one email is 6.6 ms, and processing 8,000 emails requires approximately 52.8 seconds of cumulative encryption time. RSA key exchange is the primary bottleneck, accounting for approximately 75.76% of the encryption time. ECC, parallel processing, and hardware acceleration can improve the performance of the cloud-based email encryption system.